4. 6

1 (1)

$$\lim_{x \to \frac{\pi}{2}} \left(x - \frac{\pi}{2} \right) \tan x$$

$x - \frac{\pi}{2} = t$

$$\lim_{t \to 0} t \cdot \tan\left(t + \frac{\pi}{2} \right) = \lim_{t \to 0} - \frac{t}{\tan t} = -1 \text{ //}$$

(6)

$$\lim_{x \to 0} \frac{\tan x}{\sin x + x} = \lim_{x \to 0} \frac{\sec^2 x}{\cos x + 1} = \frac{1}{2} \text{ //}$$

(7)

$$\lim_{x \to \infty} x^{\frac{1}{x}} = \lim_{x \to \infty} (1 + x - 1)^{\frac{1}{x-1} \cdot \frac{1}{x} \cdot (x-1)} = 1$$

(8)

$$\lim_{x \to \infty} \left(1 - \frac{1}{x} \right)^{\sqrt{x}}$$

$x \to \infty$

$$\lim_{x \to \infty} \ln \left(1 - \frac{1}{x}\right)^{\sqrt{x}} = \lim_{x \to \infty} \sqrt{x} \ln \left(1 - \frac{1}{x}\right) = \lim_{x \to \infty} \frac{\ln\left(1 - \frac{1}{x}\right)}{\frac{1}{\sqrt{x}}} = \lim_{x \to \infty} \frac{\ln(x-1) - \ln x}{x^{-\frac{1}{2}}}$$

로피탈 정리를 쓰면

$$\lim_{x \to \infty} \frac{\frac{1}{x-1} - \frac{1}{x}}{-\frac{1}{2} x^{-\frac{3}{2}}} = \lim_{x \to \infty} \frac{\frac{1}{x(x-1)}}{-\frac{1}{2} \cdot \frac{1}{x\sqrt{x}}} = \lim_{x \to \infty} - \frac{2\sqrt{x}}{x-1} = 0$$

$$\therefore \lim_{x \to \infty} \left(1 - \frac{1}{x} \right)^{\sqrt{x}} = 1 \text{ //}$$

2

(1)

$$\lim_{h \to 0} \frac{\tan^4 h}{h^3 \cdot \sin h \cdot \cos h} = \lim_{h \to 0} \frac{\tan^3 h}{h^3} \cdot \frac{\tan h}{\sin h} \cdot \frac{1}{\cos h}$$

$$= 1 \text{ //}$$

(6)

$$\lim_{x \to 0} \frac{\sin(\tan x)}{x} = \lim_{x \to 0} \frac{\sec^2 x \cdot \cos(\tan x)}{1} = 1 \text{ //}$$